13 $(6^{-2})^w = 6^8$

(a) Find the value of w.

$-2w = 8, \quad w = -4$

1 mark
Multiply the powers.

$w =$ .....

(1)

$\frac{(a^r)^2}{(a^t)^3}$ can be written in the form a^u

(b) Find an expression for u in terms of r and t.

Numerator power of a: $2r$

1 mark
Multiply powers inside brackets.

Denominator power: $3t$; $u = 2r - 3t$

1 mark
Subtract powers when dividing.

.....

(2)

Given that $n^{\frac{2}{3}} = 8$ and $n > 0$

(c) work out the value of n.
Give your answer in the form $a\sqrt{b}$ where a and b are integers.

$n^2 = 8^3 = 512$

1 mark
Cube both sides to get n^2.

$n = \sqrt{512} = \sqrt{(256 \times 2)}$

$n = 16\sqrt{2}$ (n is positive)

1 mark
Positive root; simplify the surd.

.....

(2)

(Total for Question 13 is 5 marks)